

Requirement Specification Document

E-Commerce Order Management Database System

Week 1 – Requirement Analysis and Customer Database Module

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Project Title	Gym Membership Management System
Week	Week 1

1. Introduction

The **Gym Membership Management System** is designed to manage gym members, trainers, membership plans, attendance, and payments efficiently. The system provides a centralized database to store member information and simplifies daily gym operations

1.1 Purpose

The purpose of this document is to define the business, functional, and database requirements of the Gym Membership Management System. It serves as a guide for designing the member database and developing the application.

1.2 Project Scope

The system manages member registration, membership plans, trainer assignments, attendance tracking, payment records, and report generation. It helps gym administrators reduce manual work and improve data accuracy

1.3 Intended Users

- **Members** – Register, renew memberships, and view membership details.
- **Trainers** – View assigned members and training schedules.
- **Administrators** – Manage members, trainers, membership plans, payments, and reports.

2. Business Requirements

The following are the key business needs the system must satisfy:

- Maintain member records securely
- Manage membership plans and renewals.
- Track member attendance
- Record payment transactions
- Assign trainers to members
- Generate reports on memberships, attendance, and revenue.

3. Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement Description
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FR-1	Generate reports on memberships, attendance, and revenue.
FR-2	The system shall manage membership plans and renewals.
FR-3	The system shall record member attendance
FR-4	The system shall assign trainers to members.
FR-5	The system shall record membership payments.
FR-6	The system shall generate attendance and payment reports.
FR-7	The system shall allow administrators to update or delete member records.
FR-8	The system shall provide secure login for administrators and trainers.

4. Non-Functional Requirements

- Performance: Member information should be retrieved within 2 seconds.
- Security: Personal and payment data must be protected.
- Reliability: The system should maintain accurate records with minimal downtime.
- Usability: The interface should be easy to use for staff and members.
- Scalability: The system should support multiple branches and thousands of members.

5. Customer Database Module – Data Requirements

Since this week's focus includes the Customer Database Module, the following entity and attributes have been identified:

Attribute	Data Type	Description
Member id	INT (Primary Key)	Unique member ID
full name	VARCHAR(100)	Member name
Gender	VARCHAR(10)	Gender
Phone number	VARCHAR(15)	Contact number
email	VARCHAR(100)	Email address
address	VARCHAR(255)	Residential address
Membership plan	VARCHAR(50)	Selected membership plan
Join date	DATE	Membership start date
Expiry date	DATE	Membership expiry date
Payment status	VARCHAR(20)	Paid / pending

6. Assumptions and Constraints

- The system uses a relational database such as MySQL.
- Only authorized users can access administrative functions.
- Each member has a unique membership ID.
- Internet access is required for cloud-based deployment.
- This document covers the Requirement Analysis and Member Database Module.

7. Conclusion

The **Gym Membership Management System** provides an efficient solution for managing gym operations, member information, attendance, memberships, and payments. The Member Database Module serves as the foundation for database design, ER diagrams, SQL implementation, and future system development.